

GALLBLADDER

Regular anatomy notes with enlarged source-PDF diagrams and exam-oriented explanations

Core identity

The gallbladder is an elongated pear-shaped sac attached to the inferior surface of the right lobe of the liver.

- Capacity: about 30-50 ml.
- It belongs to the extrahepatic biliary apparatus.
- It communicates with the biliary duct system through the cystic duct.

Main functions

The gallbladder stores, concentrates and expels bile according to digestive need.

- Stores bile between meals.
- Concentrates bile by absorption of water and electrolytes.
- Contracts when fat enters the duodenum.
- Emptying is mainly under the influence of cholecystokinin (CCK).

Why this topic is clinically important

The gallbladder is frequently involved in cholecystitis, cholelithiasis, obstructive jaundice and surgical procedures like cholecystectomy. The anatomy of the cystic duct, cystic artery and Calot triangle is especially important because errors in this region can cause bleeding or bile duct injury.

LOCATION, DIMENSIONS AND PARTS

Location

The gallbladder lies in the fossa for the gallbladder on the inferior surface of the right lobe of the liver, along the right edge of the quadrate lobe.

- It extends from the right end of the porta hepatis to the inferior border of the liver.
- It is usually attached to the liver by connective tissue.
- It may rarely be suspended by a short mesentery or partly buried in the liver.

Dimensions and parts

Length is about 10 cm and width is about 3 cm at the widest part.

- Parts: fundus, body and neck.
- Fundus: expanded blind end.
- Body: main part between fundus and neck.
- Neck: narrow upper end continuing as cystic duct.

Specimen and diagram orientation: In a liver specimen, first identify the porta hepatis and quadrate lobe, then trace the gallbladder from the inferior border of the liver towards the right end of the porta hepatis. The fundus is the rounded projecting end, while the neck is narrow and continues as the cystic duct. This orientation prevents confusion between the gallbladder, hepatic ducts and nearby vessels.

- Fundus points downwards and forwards.
- Neck lies near the porta hepatis and cystic duct.
- Upper surface is attached to liver; lower surface is peritoneal.

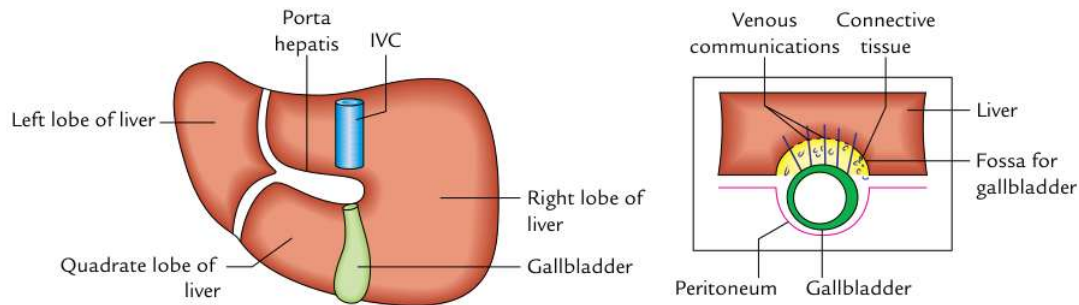
Fig. 8.14 - Location of the gallbladder on the inferior surface of liver

Fig. 8.14 Location of the gallbladder on the inferior surface of the right lobe of the liver. Figure in the inset on the right side shows connective tissue and venous communication between gallbladder and liver.

This diagram fixes the position of the gallbladder in relation to the porta hepatis, quadrate lobe and inferior surface of the liver. The inset explains why the gallbladder is closely attached to the liver: connective tissue and venous communications link the gallbladder bed with the liver substance.

Part	Main description	Important relations
Fundus	Expanded blind end projecting below the inferior border of liver.	Anterior abdominal wall at the tip of right 9th costal cartilage; posteriorly transverse colon.
Body	Directed upward, backward and to the left towards the neck.	Upper surface related directly to liver; undersurface related to the second part of duodenum.
Neck	Narrow upper end near the right end of porta hepatis; continues as cystic duct.	Inferiorly related to first part of duodenum; cystic artery lies in loose areolar tissue near it.

RELATIONS AND SURFACE ANATOMY

Relationship map: The gallbladder should be learned by surfaces. The hepatic surface is fixed to the liver, while the free inferior surface is peritoneal and related to abdominal viscera. The fundus reaches the anterior abdominal wall, the body lies against the duodenum and colon, and the neck lies near the porta hepatis. This surface-based method makes relations easier to recall in practical exams.

- Fundus: anterior abdominal wall and transverse colon.
- Body: liver above, duodenum and colon below.
- Neck: close to porta hepatis, cystic duct and cystic artery.

Clinical use of relations: Inflammation of the gallbladder may form adhesions with neighbouring organs, especially the duodenum and colon. A stone in Hartmann pouch can press near the cystic duct or common hepatic duct region, so relations are not only descriptive but also explain pain, obstruction and surgical difficulty.

- Duodenal relation explains possible cholecystoenteric fistula.
- Hepatic attachment explains the gallbladder bed during surgery.
- Anterior wall contact explains Murphy sign.

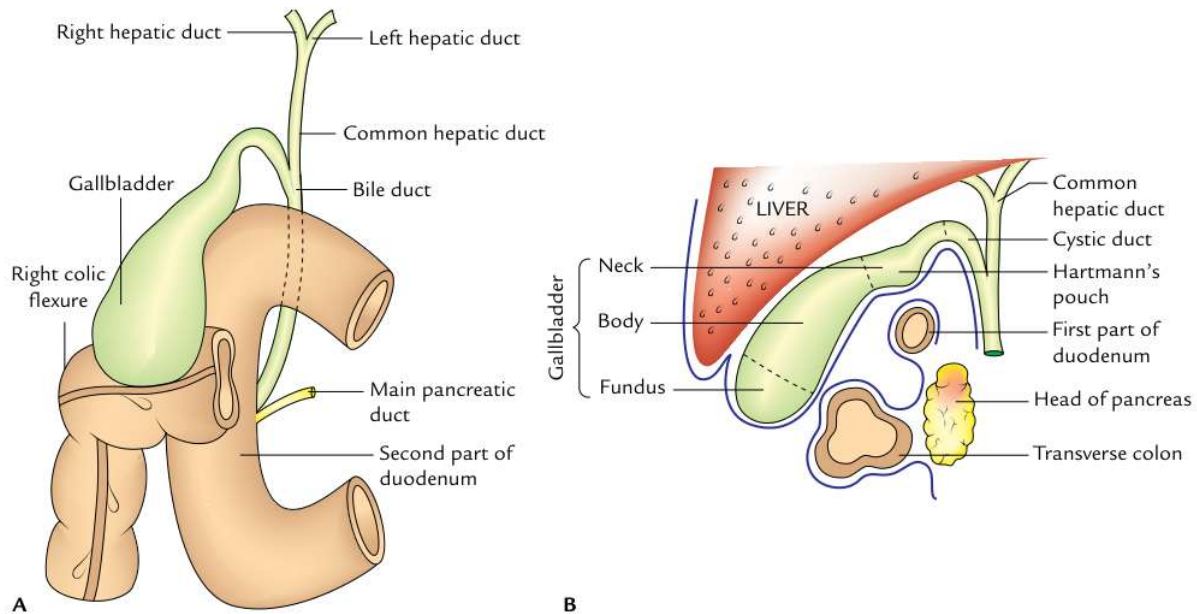
Fig. 8.15 - Relations of the gallbladder

Fig. 8.15 Relations of the gallbladder: **A**, as seen from the front after removal of the liver; **B**, as seen in sagittal section through the gallbladder fossa.

The front view shows the gallbladder connected to the duct system and related to the duodenum, pancreas and transverse colon. The sagittal view shows the fundus, body and neck in the gallbladder fossa. Notice Hartmann pouch at the neck, because stones lodged here may lead to inflammation or adhesion to the first part of the duodenum.

Surface marking of fundus

The fundus touches the anterior abdominal wall at the tip of the right 9th costal cartilage.

- This point is deep to the place where the right linea semilunaris meets the costal margin.
- Tenderness here is important in acute cholecystitis.

Hartmann pouch

Hartmann pouch is a pouch-like dilatation of the posteromedial wall of the neck.

- It is directed downward and backward.
- Gallstones may lodge here.
- Stones may cause adhesion with the first part of the duodenum and can rarely perforate it.

STRUCTURE OF GALLBLADDER WALL

Layers of the wall

The gallbladder wall is built for storage, concentration and contraction of bile. It lacks the typical muscularis mucosae and submucosa pattern seen in many hollow viscera. Its mucosa forms minute folds and has tall columnar epithelium.

- Serous layer of peritoneum covers the free surface.
- Subserous layer consists of loose areolar tissue.
- Fibromuscular layer contains fibrous tissue mixed with smooth muscle fibres arranged in longitudinal, circular and oblique bundles.
- Mucous membrane is thrown into minute rugae giving a honeycomb appearance; epithelium is simple tall columnar.

Peritoneal relation

The undersurface of the gallbladder is covered by peritoneum.

- The upper surface is attached to the liver and is devoid of peritoneum.
- Because of this, infection can spread between the gallbladder and liver bed.

Functional meaning

Concentration of bile occurs by absorption through the mucosa.

- Smooth muscle contracts during gallbladder emptying.
- CCK causes contraction of the gallbladder and relaxation of the sphincters at the distal bile passage.

High-yield structural interpretation

The anatomical arrangement of the gallbladder wall explains both its normal function and clinical behavior. The mucosa helps concentrate bile, the fibromuscular coat expels bile during digestion, and the attached hepatic surface explains why inflammation or venous spread can involve the gallbladder bed of the liver. In exams, connect the wall structure with bile concentration, contraction after fatty meals and spread of inflammation.

- No submucosa is classically described in the gallbladder wall.
- The mucosa is thrown into fine folds, creating a honeycomb appearance.
- The non-peritoneal hepatic surface is the surgical bed during cholecystectomy.
- Pain is often made worse after fatty food because contraction increases pressure against an obstructing stone.

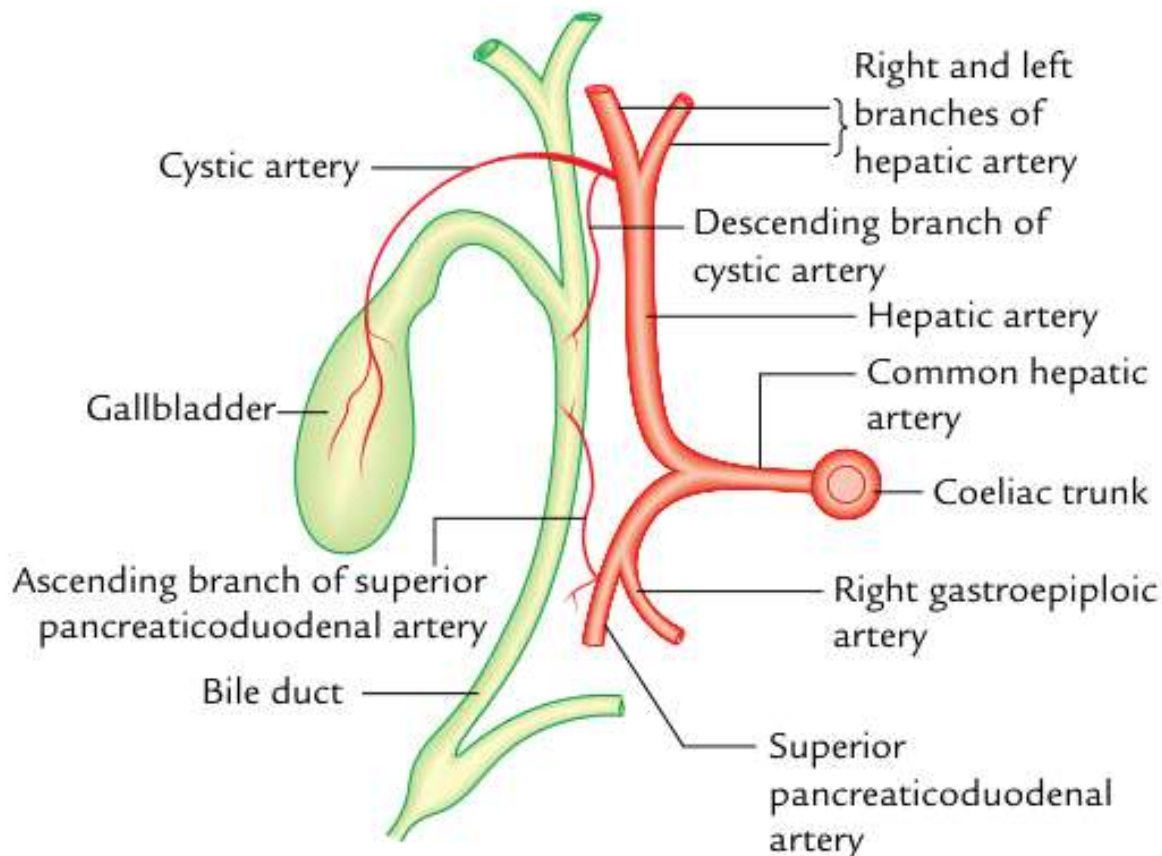
BLOOD SUPPLY, VENOUS DRAINAGE, LYMPH AND NERVE SUPPLY

Orientation before the supply diagram

The vascular, lymphatic and neural supply of the gallbladder follows its close relationship with the liver, cystic duct and lesser omentum. The cystic artery is the main artery to identify surgically, while cystic lymph node of Lund is an important marker in inflammation of the gallbladder.

Before reading the supply diagram: Remember that the gallbladder is not supplied like an isolated organ. Its artery, veins, lymphatics and nerves are closely related to the hepatic hilum and the extrahepatic biliary ducts. The cystic artery is usually found within Calot triangle, so its variation is a practical surgical problem rather than just a theoretical point.

- Cystic artery = chief arterial supply.
- Cystic vein and small gallbladder-bed veins explain spread to liver.
- Cystic lymph node may enlarge in cholecystitis.

Fig. 8.16 - Blood supply of the gallbladder and bile duct

The cystic artery is the chief artery of the gallbladder and usually arises from the right hepatic artery. The diagram also shows the blood supply around the bile duct. This is surgical anatomy: correct identification of the cystic artery prevents bleeding during cholecystectomy.

Supply/drainage	Details to remember
Arterial supply	Mainly by cystic artery, usually a branch of the right hepatic artery. It may also arise from the main hepatic artery, left hepatic artery or gastroduodenal artery.
Venous drainage	By cystic vein to portal vein and by small veins passing from the superior surface of gallbladder into the liver through the gallbladder bed, draining finally into hepatic veins.
Lymphatic drainage	Mainly to cystic lymph node of Lund in Calot triangle and to nodes along the upper part of bile duct; finally to coeliac lymph nodes. Few vessels communicate with subcapsular lymphatics of liver.
Nerve supply	Via cystic plexus: sympathetic fibres T7-T9, parasympathetic fibres from vagus nerves, and fibres from right phrenic nerve.

Referred pain pattern

Gallbladder pain may be referred to the inferior angle of the right scapula through sympathetic fibres, to the tip of the right shoulder through the right phrenic nerve, and sometimes towards the stomach through vagal pathways. This explains why gallbladder disease may be felt away from the right hypochondrium.

CYSTIC DUCT AND SPIRAL VALVE OF HEISTER

Cystic duct

The cystic duct is about 3-5 cm long.

- It runs backward and downward from the neck of the gallbladder.
- It joins the common hepatic duct at an acute angle to form the bile duct/common bile duct.
- The junction usually lies just below the porta hepatis.

Spiral valve of Heister

The mucosa of the cystic duct forms 5-10 crescentic folds.

- These folds project into the lumen in a spiral fashion.
- Together they form the spiral valve of Heister.
- This helps keep the cystic duct open for movement of bile in and out of the gallbladder.

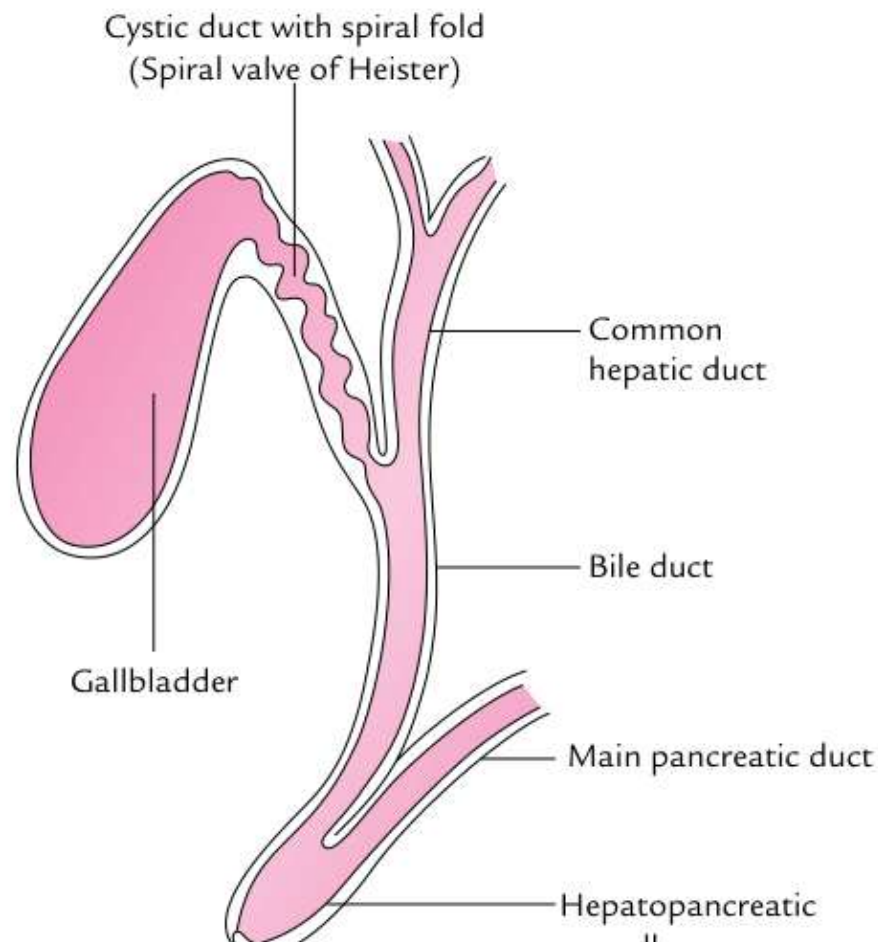
Flow of bile through the cystic duct

When the distal bile passage is closed, bile secreted by the liver fills the hepatic ducts and passes through the cystic duct into the gallbladder for storage and concentration. When the distal passage opens during digestion, bile flows into the duodenum from the common hepatic duct and from the contracting gallbladder. This bidirectional flow explains why the cystic duct needs to remain patent rather than behaving like a one-way valve.

- Between meals: bile is diverted into the gallbladder.
- After fatty meal: CCK contracts the gallbladder and relaxes distal sphincters.
- Obstruction of the cystic duct can distend the gallbladder and produce colicky pain.

Viva-friendly wording

Say this in exams: The cystic duct is short, tortuous and contains spiral mucosal folds called the spiral valve of Heister. It connects the neck of the gallbladder with the common hepatic duct and allows bile to pass both into and out of the gallbladder. Its close relationship with the common hepatic duct and cystic artery makes it important in cholecystectomy.

Fig. 8.17 - Spiral valve of Heister in cystic duct

This enlarged diagram shows how the cystic duct connects the neck of the gallbladder with the common hepatic duct. The spiral fold is not a true valve that stops flow; rather, it maintains duct patency and helps bile pass both towards and away from the gallbladder.

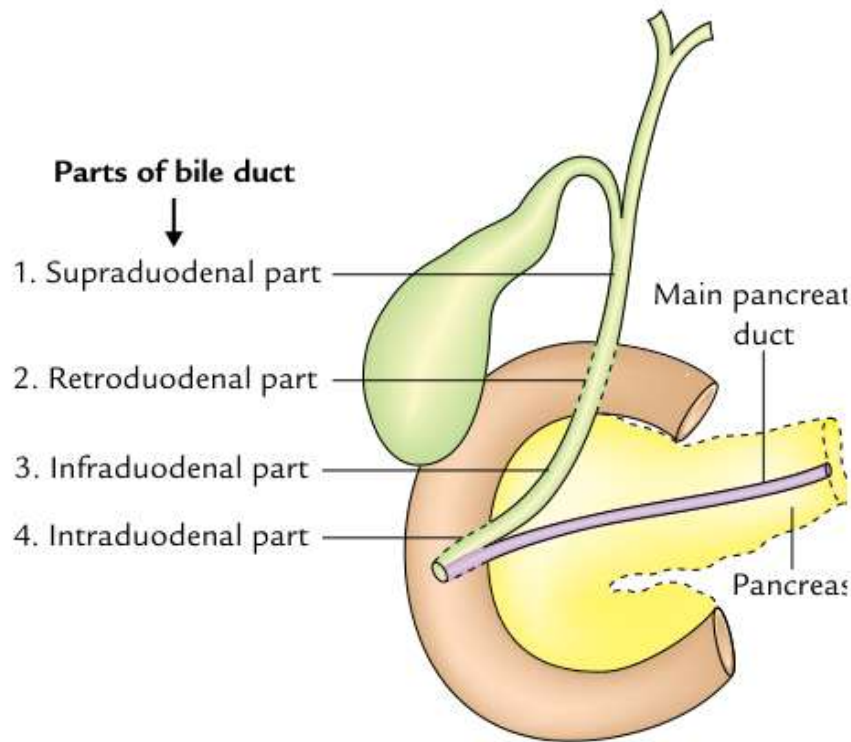
BILE DUCT / COMMON BILE DUCT (CBD)

Formation and basic course

The bile duct is formed near the porta hepatis by the union of the cystic duct and the common hepatic duct. It is usually about 7.5 cm long and about 6 mm in diameter. It carries bile from the liver and gallbladder towards the second part of the duodenum.

How to learn the CBD course: Follow the duct from the porta hepatis to the duodenum in four steps: first in the free margin of the lesser omentum, then behind the first part of duodenum, then near or within the posterior surface of pancreas, and finally through the wall of the second part of duodenum. These steps also explain common sites of obstruction and compression.

- Supraduodenal part is important in the portal triad.
- Retroduodenal and pancreatic parts are affected by duodenal/pancreatic disease.
- Intraduodenal part joins the main pancreatic duct near the major duodenal papilla.

Fig. 8.18 - Parts of the bile duct**Fig. 8.18** Parts of the bile duct (dark green colour).

The bile duct is divided into four parts: supraduodenal, retroduodenal, infraduodenal/pancreatic and intraduodenal. This arrangement is important because stones, pancreatic disease or surgical injury may involve different parts of the duct.

Part of CBD	Course and relations
Supraduodenal part	Descends in the right free margin of the lesser omentum; lies to the right of hepatic artery proper and anterior to portal vein.
Retroduodenal part	Descends behind the first part of the duodenum; gastroduodenal artery lies on its left and IVC posteriorly.
Infraduodenal/pancreatic part	Runs in a groove on the posterior surface of pancreas; may be embedded in pancreatic tissue.
Intraduodenal/intramural part	Enters the posteromedial wall of the second part of duodenum and unites with the main pancreatic duct to form the hepatopancreatic ampulla.

ACCESSORY HEPATIC DUCTS AND CALOT TRIANGLE

Surgical relevance of variations: Accessory hepatic ducts and arterial variations are important because the biliary tree is commonly approached during cholecystectomy. A small accessory duct may be missed if only the cystic duct is searched for, and the right hepatic artery may form a loop close to the cystic artery. This is why surgeons carefully identify the critical structures in Calot triangle before clipping.

- Look for duct variations before ligating.

- Do not assume every duct entering the gallbladder region is the cystic duct.
- Arterial loops may mimic the cystic artery.

Fig. 8.13 - Sites of termination of accessory hepatic ducts

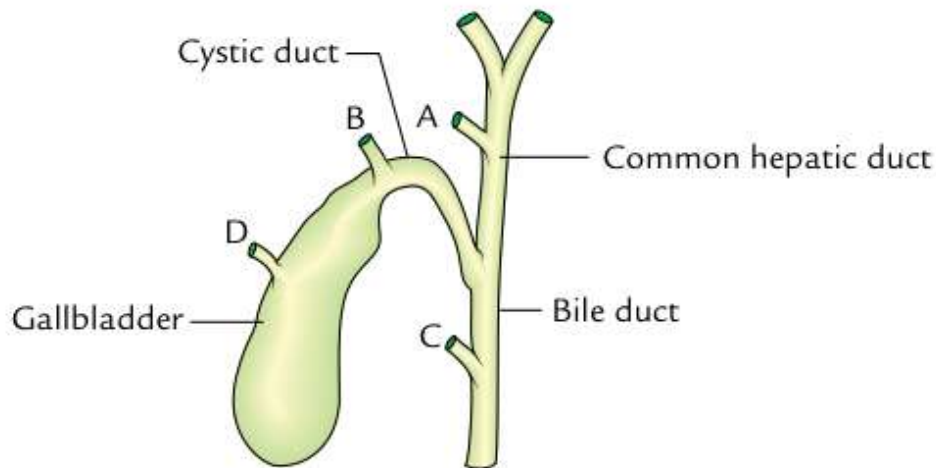
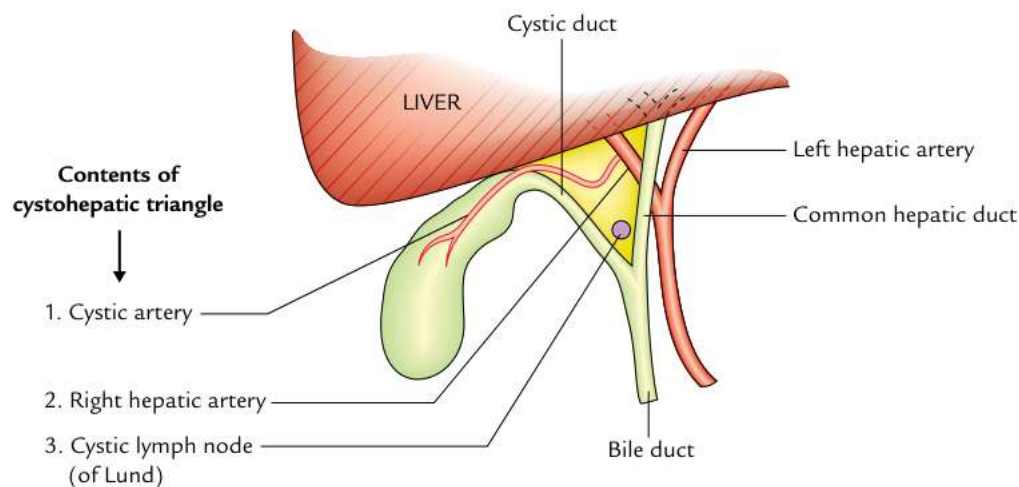


Fig. 8.13 Sites of termination of the accessory hepatic ducts.

Accessory hepatic ducts are seen in a minority of cases and often arise from the right lobe. They may open into the common hepatic duct, cystic duct, bile duct or gallbladder. This variation matters during cholecystectomy because injury to these ducts can cause postoperative bile leakage.

Fig. 8.19 - Cystohepatic triangle of Calot



Calot triangle is bounded by the cystic duct on the right, common hepatic duct on the left, and the inferior surface of the liver above. Its important contents are the cystic artery, right hepatic artery and cystic lymph node of Lund. This triangle is the key surgical landmark for safe removal of the gallbladder.

Boundary/content	Details
Right boundary	Cystic duct.

Left boundary	Common hepatic duct.
Superior boundary	Inferior surface of the liver.
Contents	Cystic artery, right hepatic artery and cystic lymph node of Lund.
Importance	Helps locate the cystic pedicle during cholecystectomy; variations of ducts and arteries are common here.

Fig. 8.20 - Moynihan's hump

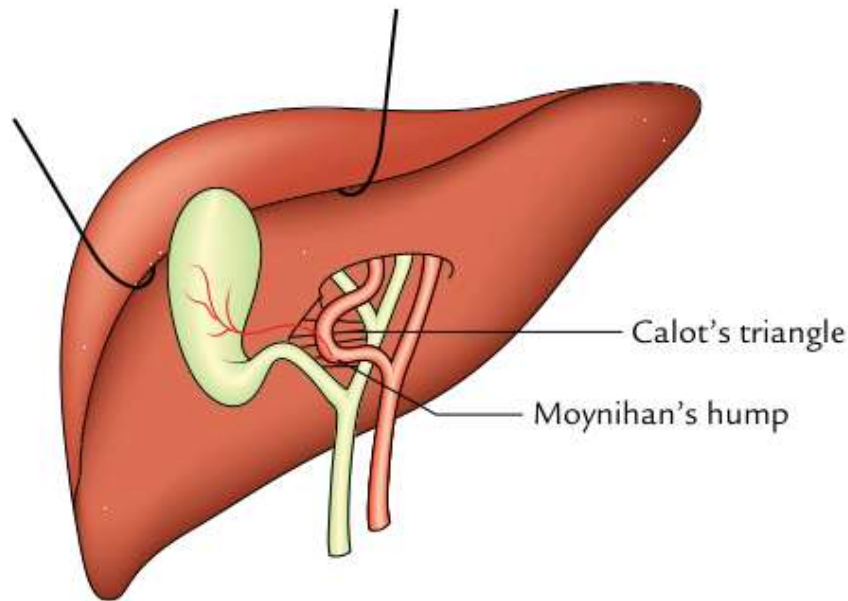


Fig. 8.20 Moynihan's hump.

Moynihan's hump is a tortuous caterpillar-like loop of the right hepatic artery near the cystic artery. It is dangerous during surgery because it can be mistaken for the cystic artery and accidentally clamped or cut, producing profuse bleeding.

APPLIED ANATOMY

Cholecystitis

Cholecystitis is inflammation of the gallbladder. In acute cholecystitis there may be sudden pain in the right hypochondrium, vomiting and tenderness. Murphy sign is positive when pressing under the right costal margin at the tip of the 9th costal cartilage during deep inspiration produces sharp pain and a catch in breath.

Murphy sign explained anatomically: During deep inspiration, the liver and gallbladder descend. If an inflamed gallbladder touches the examiner's fingers placed at the tip of the right 9th costal cartilage, sudden pain stops inspiration. This links surface anatomy directly with clinical examination.

- Positive sign suggests acute cholecystitis.
- Pain is located near the right costal margin.
- Respiratory arrest occurs because the inflamed gallbladder contacts the palpating fingers.

Cholelithiasis and biliary colic

Cholelithiasis means gallstone formation.

- Chronic cholecystitis commonly leads to gallstones.
- Biliary colic is caused by spasm of smooth muscle attempting to expel a stone.
- Pain is intermittent and severe when a stone is impacted at the cystic duct or lower bile duct.

Obstructive jaundice

Obstruction of CBD may occur due to gallstones, carcinoma of head of pancreas or enlarged neoplastic hepatic lymph nodes.

- Obstruction blocks bile flow.
- This leads to jaundice.
- A palpable enlarged gallbladder with obstructive jaundice suggests extrinsic obstruction more than stone obstruction.

Courvoisier law

Obstructive jaundice with a distended, palpable gallbladder is most likely due to extrinsic obstruction of the CBD, such as carcinoma of the head of pancreas. In contrast, obstruction by gallstones often occurs with a contracted, fibrotic gallbladder due to previous cholecystitis, so the gallbladder is usually not distended.

IMAGING OF THE BILIARY TRACT

Technique	Core idea and use
Cholecystography	Radiopaque dye is given orally or intravenously, excreted by liver into bile, concentrated in gallbladder and then visualized by X-ray.
Ultrasonography	Noninvasive method for visualizing gallbladder and biliary apparatus; now largely replaces cholecystography.
ERCP	Endoscope is used to cannulate the hepatopancreatic ampulla and inject contrast to visualize bile and pancreatic ducts.
MRI/MRCP	Noninvasive magnetic resonance imaging can demonstrate bile and pancreatic ducts without endoscopic cannulation.

Imaging interpretation note: On ultrasound, gallstones may appear as bright echoes with acoustic shadowing, while dilatation of the bile duct suggests obstruction distal to the cystic duct. ERCP can both outline the ducts and allow therapeutic removal of stones in selected cases, whereas MRCP gives a noninvasive map of the biliary tree.

- Ultrasound is the common first investigation.
- ERCP is invasive but may be diagnostic and therapeutic.
- MRCP is noninvasive and useful for duct mapping.

Exam tip

For viva and short notes, always connect gallbladder anatomy with Calot triangle, cystic artery, referred pain, Murphy sign, gallstones and obstructive jaundice. These are the high-yield clinical correlations.

QUICK REVISION CHART

Question	Answer
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Capacity of gallbladder	30-50 ml.
Location	Fossa for gallbladder on inferior surface of right lobe of liver, near quadrate lobe.
Parts	Fundus, body and neck.
Surface point of fundus	Tip of right 9th costal cartilage where right linea semilunaris meets costal margin.
Main artery	Cystic artery, usually from right hepatic artery.
Lymph node in Calot triangle	Cystic lymph node of Lund.
Nerve supply	Cystic plexus: sympathetic T7-T9, vagus fibres and right phrenic fibres.
Referred shoulder pain pathway	Irritation of diaphragmatic peritoneum via phrenic nerve C3-C5.
Cystic duct mucosal fold	Spiral valve of Heister.
Hormone causing gallbladder contraction	Cholecystokinin (CCK).

One-paragraph summary

The gallbladder is a pear-shaped bile reservoir situated on the inferior surface of the right lobe of the liver. It has fundus, body and neck, drains through the cystic duct, and receives arterial supply mainly from the cystic artery. Its lymph drains to the cystic node of Lund and coeliac nodes, and its nerves reach through the cystic plexus. Clinically, the most important points are gallstones, cholecystitis, Murphy sign, referred pain to the right shoulder/scapular region, Calot triangle and safe identification of the cystic artery during cholecystectomy.

Answer-writing sequence: For a regular anatomy answer, write in the order: definition, situation, parts, relations, wall, peritoneal covering, blood supply, lymph drainage, nerve supply, cystic duct, bile duct and applied anatomy. This order prevents missing small but scoring points, especially supply and clinical correlations.

- Do not skip surface marking of fundus.
- Always mention cystic artery and cystic lymph node.
- Add Calot triangle as a separate labelled point.

Important specimen points: In practical anatomy, identify the fundus first because it projects beyond the inferior border of liver. Then follow the body to the neck and cystic duct. The gallbladder bed on the liver explains why venous channels and lymphatics can communicate with liver tissue.

- Fundus = rounded blind end.
- Neck = narrow part continuing into cystic duct.
- Gallbladder bed = non-peritoneal hepatic surface.

How to frame a 5-mark answer

Start with definition and location, then write parts and relations, followed by structure and supply. Add cystic duct with spiral valve of Heister, then finish with applied anatomy: cholecystitis, cholelithiasis, Murphy sign, referred pain and Calot triangle. Keep one labelled diagram of the gallbladder/CBD or Calot triangle in the answer for better scoring.

- Opening line: pear-shaped sac in fossa for gallbladder on inferior surface of liver.
- Middle: parts, relations, wall, cystic artery, lymph and nerves.
- Clinical ending: stones, inflammation, obstructive jaundice and surgical triangle.

Common mistakes to avoid

Do not confuse cystic duct with common bile duct.

- Do not forget the cystic lymph node of Lund.
- Do not write that the spiral valve blocks bile flow; it maintains patency.
- Do not omit the right 9th costal cartilage surface point.

High-yield clinical links

Stone in cystic duct causes gallbladder distension and colicky pain.

- Stone in common bile duct causes obstructive jaundice.
- Pancreatic head carcinoma can compress the lower CBD.
- Inflamed gallbladder causes Murphy sign and referred pain.

Last-minute viva checklist

Before closing revision, confirm these seven points aloud: pear-shaped sac, inferior surface of right lobe of liver, fundus at right 9th costal cartilage, three parts, cystic duct with spiral valve of Heister, cystic artery from right hepatic artery, and Calot triangle containing cystic artery, right hepatic artery and cystic lymph node.

Final keywords to remember

Gallbladder bed, Hartmann pouch, spiral valve of Heister, cystic artery, cystic lymph node of Lund, Calot triangle, Murphy sign, biliary colic, obstructive jaundice and Courvoisier law are the most exam-repeated words from this topic. Use these terms in your short notes and viva answers to make the answer sound complete.